

Scientists investigated the effect of a heat treatment on mass transport in barley plants.

- They applied steam to one short section of a leaf of the heat-treated plants. This area is shown by the arrows in **Figure 9**.
- They did not apply steam to the leaves of control plants.
- They then supplied carbon dioxide containing radioactively-labelled carbon to each plant in the area shown by the rectangular boxes in **Figure 9**.
- After 4 hours, they:
 - found the position of the radioactively-labelled carbon in each plant. These results are shown in **Figure 9**.
 - recorded the water content of the parts of the leaf that were supplied with radioactively-labelled carbon dioxide. These results are shown in **Table 4**.

Figure 9

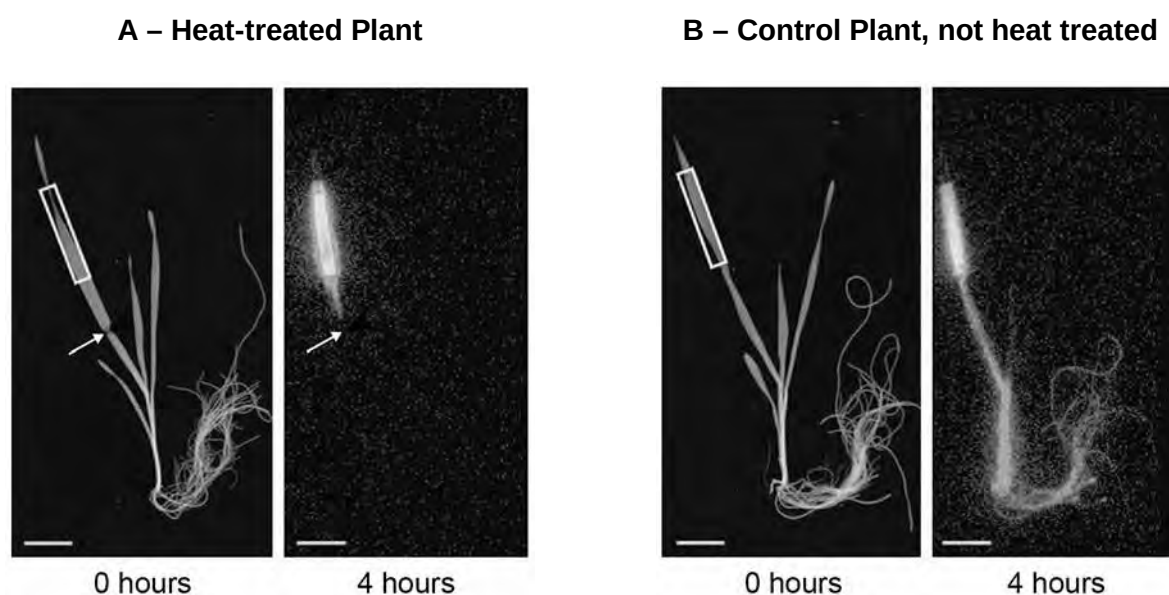


Table 4

Plant from which the leaf was taken	Water content of leaf / % of maximum (± 2 standard deviations)
Heat-treated Plant A	84.6 (± 11.3)
Control Plant, not heat treated B	92.8 (± 8.6)

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The scientists concluded that this heat treatment damaged the phloem.

Explain how the results in **Figure 9** support this conclusion.

[2 marks]

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The scientists also concluded that this heat treatment did not affect the xylem.

Explain how the results in **Table 4** support this conclusion.

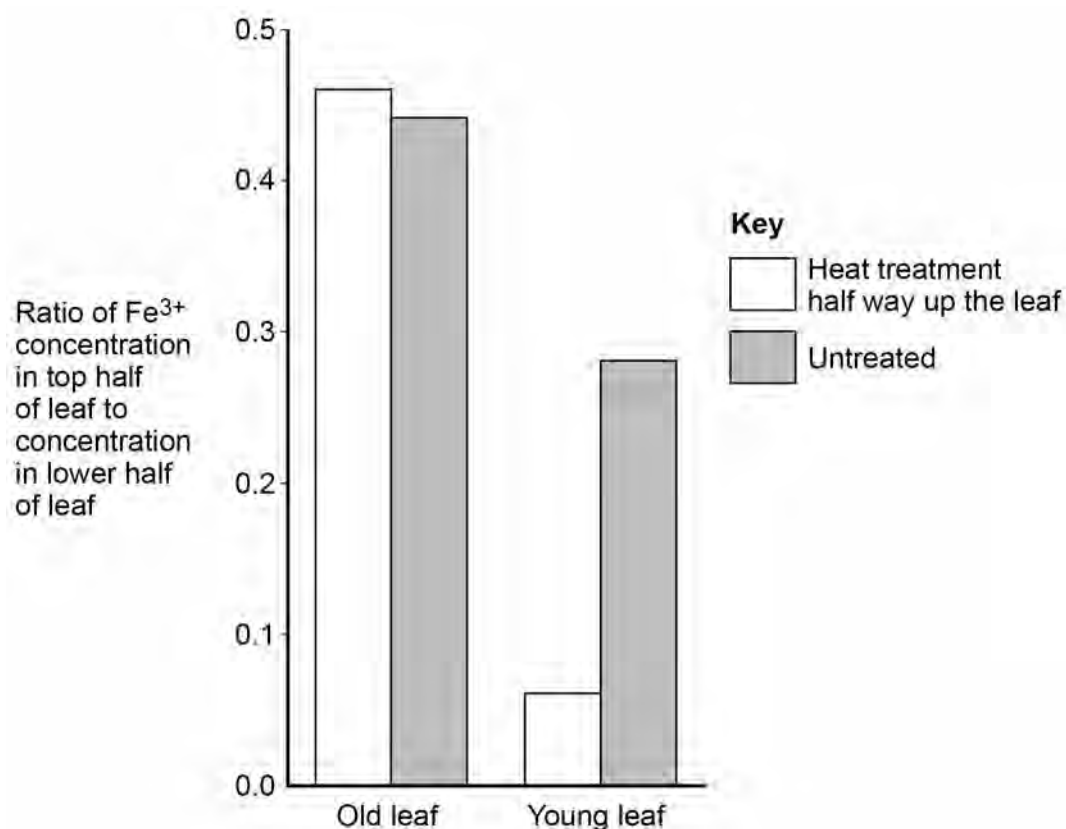
[2 marks]

0 8 . 3

The scientists then investigated the movement of iron ions (Fe^{3+}) from the soil to old and young leaves of heat-treated barley plants and to leaves of plants that were not heat treated. Heat treatment was applied half way up the leaves. The scientists determined the concentration of Fe^{3+} in the top and lower halves of the leaves of each plant.

Their results are shown in **Figure 10**.

Figure 10



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